

David Zhang

dzhang17@mit.edu

<https://www.linkedin.com/in/dzhang17>

<https://heyo392.github.io>

EDUCATION

Massachusetts Institute of Technology

Sep 2023 – May 2027

Bachelor of Science in Computer Science, Physics (GPA: 5.00 / 5.00)

- **Relevant Coursework:** Inference and Information (G), Machine Learning (G), Multi-agent Learning (G), Design and Analysis of Algorithms, Data Structures and Algorithms, Computation Structures, Software Construction, Quantum Mechanics II

EXPERIENCE

Amazon

Jun 2025 – Aug 2025

Software Development Engineer Intern

- Designed and implemented a new Kotlin package for Kindle on Android.
- Architected MVVM modules and service layer; built Compose UI; wrote Robolectric tests; established internal CI/CD and a standalone test app.
- Integrated with company SDKs for data fetching, updating the host app to adopt the package.

International Olympiad on Astronomy and Astrophysics

Aug 2022, Aug 2023

Competitor

- Represented team USA at the International Olympiad for 2022 and 2023, awarded Gold and Silver respectively.
- Placed 2nd on the National Astronomy Competition, the national level qualifying exam.

MIT CSAIL - Pulkit Agrawal's Lab

Sep 2025 – Present

Undergraduate Researcher (UROP)

- Conducting research in memory systems for robotic Imitation Learning and Reinforcement Learning.

MIT Plasma Science Fusion Center

Sep 2024 – May 2025

Undergraduate Researcher (UROP)

- Modified existing active learning pipeline to predict properties of the pedestal within a Tokamak fusion reactor.
- Trained Bayesian networks on the EPED simulation, using uncertainty outputs to inform future experiments.
- Gained a background in plasma physics and ML packages, extended an active learning pipeline, generated datasets, and tested the model.

PROJECTS

MIT Pokerbots 2026 1st Place, \$10K Prize | C++, Python

- Built a poker AI from scratch using Counterfactual Regret Minimization to approximate a Nash Equilibrium for a custom 3-card Hold'em variant.
- Implemented equilibrium refinements that significantly improved robustness, contributing to a 96% winrate.
- Trained the final strategy over 3 trillion hands at 10M hands/sec on 128 threads.
- Achieved 1st place out of 80 teams, losing half as many games as 2nd place.

Team Leduc Poker (Graduate Multi-Agent Learning Final Project) | C++, Python

- Implemented a high-performance Team-Belief DAG DCFR solver and designed a game-theory-guided compression scheme to approximate the full TB-DAG equilibrium with a 100-tree correlated strategy ensemble for 4-player Team Leduc Hold'em.
- Achieved best-in-class exploitability (over 60× lower than the instructor baseline, 3× lower than the next-best team) and an undefeated record with the highest final PnL (+26.00 chips vs. 22 competitors).

HONORS

IOAA Gold Medalist (2022), Science Olympiad 2x National Team Champion (2021, 2022), M3C Mathworks Technical Computing Finalist (2023), MIT Pokerclub Tournament Placings: 2/80 (Fall 2023), 3/100 (Spring 2024), Mason Science Olympiad Club Captain (2023)

SKILLS

Programming Python, C++, TypeScript, Kotlin

Tools/Frameworks Android Studio, Git, Jetpack Compose, Pandas, NumPy, PyTorch, Matplotlib, Adobe Lightroom